

AFONSO KEVIN BURGINSKI

Senior Full-Stack Engineer — Distributed Systems · Video Streaming · NestJS / Next.js

Email: afonsoburginski@gmail.com • Portfolio: afonsodev.com • GitHub: github.com/afonsoburginski • LinkedIn: linkedin.com/in/afonsoburginski

SUMMARY

Senior Full-Stack Engineer with 7 years of market experience architecting and operating distributed systems end to end. Currently owns the streaming domain of a large-scale smart-city video monitoring platform — WebRTC, LL-HLS, real-time analytics and event-driven microservices on Kubernetes. Deep specialist in video streaming and platform engineering; staff-track scope with full ownership from system design to production.

SKILLS

****Backend & Microservices:**** NestJS, Node.js, Go, Kotlin (Ktor), Kafka, Redis, BullMQ, Kong, CQRS, Event-Driven Architecture, REST, WebSockets

****Data:**** PostgreSQL (time partitioning, zero-downtime indexing), Prisma, Drizzle, Supabase

****Video & Streaming:**** FFmpeg/NVENC, WebRTC, HLS/LL-HLS, ABR, H.264/H.265, ONVIF, LiveKit, Cloudflare R2/CDN

****Frontend:**** Next.js (App Router, RSC), React, Angular, TypeScript, TailwindCSS, Zustand, TanStack Query

****Quality & Observability:**** Testcontainers, Jest, Prometheus, Pino structured logging

****DevOps & Cloud:**** Docker, Kubernetes (RKE2), Helm, Terraform, GitHub Actions, Nx monorepo (remote cache), AWS, Vercel

EXPERIENCE

Senior Software Engineer — Atman Systems

Mar 2026 – Present | Brazil

- Owns the streaming domain of a large-scale smart-city video platform: WebRTC with LL-HLS fallback (~250ms), dynamic H.265/H.264 codec negotiation per-client and adaptive quality via native camera substreams — infrastructure cost per camera, not per viewer.
- Built the real-time video analytics pipeline: per-frame detections on edge devices, on-device Kafka producers and an idempotent reconciler that restores analytics state after device reboots.

- Designed multi-camera event correlation (sliding-window clustering, UUIDv5 alarm dedup, auto-resolution) and telemetry at scale — time-partitioned PostgreSQL, Prometheus metrics, zero-downtime indexing.
- Delivered the platform's notification gateway from scratch: email, WhatsApp, WebSocket and push via BullMQ (exponential backoff, DLQ), multi-locale Handlebars templates and full audit trail.
- Established CI integration-test gate (Testcontainers), remote Nx cache and production Kubernetes provisioning (RKE2 + Terraform + Helm); 75+ merged PRs in 5 months. Selected to lead the platform's upcoming mobile application.

Full Stack Engineer — JHONROB Silos e Secadores

Nov 2023 – Mar 2026 | Brazil

- Led GEM ERP (Next.js + Supabase/PostgreSQL): reduced commercial proposal errors by 80%, increased monthly profit ~14%, 100+ daily active users — real-time multi-view Kanban, corporate chat, GPT-powered assistant and append-only audit trail.
- Engineered Kotlin Multiplatform tooling (Compose + Ktor + SQLDelight) for automated CAD export workflows and Android field-ops modules; multi-layer caching with sub-200ms real-time sync via Docker/Nginx VPS.

Senior Full-Stack Engineer — afonsodev.com (Founder & Independent Consultant)

Jan 2020 – Present | Remote

Stormzplus — Subscription VOD Platform (Founder & Lead Engineer)

- Full-stack streaming platform: Next.js + Supabase, GPU-accelerated Go/FFmpeg/NVENC transcoding (240p–4K HLS), Cloudflare R2/CDN, Stripe + Mercado Pago PIX subscriptions with idempotent webhooks.
- LiveKit Watch Party (up to 50 concurrent viewers) with custom signaling over data channels; custom HLS player with captions, resume, quality selector and background playback.

Orça Norte — B2B Construction Marketplace (orcanorte.com.br)

- SSR/ISR marketplace for 500+ companies: multi-tenant vendor system, subscription plans, RBAC and operational dashboards. Next.js, Prisma, PostgreSQL, NextAuth.

Full Stack Developer — TopSapp

Jan 2021 – Sep 2022 | Brazil

- ISP management interfaces (Next.js/React); legacy system migration; GitLab CI/CD; Scrum/Kanban delivery.

System Developer — Ecocentauro Sistemas Inteligentes

Jan 2019 – Oct 2020 | Brazil

- PostgreSQL schema design, query optimization and financial/operational reporting.

OPEN SOURCE

Next.js FFmpeg Transcoder (MIT) · github.com/afonsoburginski/nextjs-ffmpeg-transcoder

Go video ingestion service with FFmpeg/NVENC GPU acceleration, HLS transcoding and Cloudflare R2 storage.

EDUCATION & CERTIFICATIONS

B.Sc. Information Systems — Unemat Sinop · 2020–2024

Meta Back-End Developer Professional Certificate · IBM Full Stack Software Developer Professional Certificate

LANGUAGES

Portuguese: Native · **English:** B2 — technical communication and client meetings